

Auteur: Luca Letroye
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$$\int (e^x + 1)^2 dx$$

$$t = e^x + 1 \Rightarrow e^x = t - 1$$

$$\frac{dt}{e^x}$$

$$\Rightarrow \int \frac{t^2}{t-1} dt$$

$$= \int \left(\frac{t^2 - 1 + 1}{t-1} \right) dt$$

$$= \int \left(\frac{(t-1)(t+1) + 1}{t-1} \right) dt$$

$$= \int \left((t+1) + \frac{1}{t-1} \right) dt$$

$$\frac{1}{2}t^2 + t + \ln|t-1| + C$$

$$\frac{1}{2}(e^x + 1)^2 + (e^x + 1) + \ln|e^x| + C$$

References